Crossover filter schematics

Redrawn from LspCAD 6 GenFilter files in ?Test av olika nya filter?

Page 2: optimized_filter_1_std_comp.lsp

Bass: L1 3.3 mH?{C2 18 µF?com}?{6.8 ?+33 µF?com}

Mid: L1 0.68 mH?C1 27 µF?{C2 15 µF?com}?{L2 2.2 mH?com}?(0.15 mH ? 6.8 µF ? 22 ?)?{6.8 ?+15 µF?com}

Treble: C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{5.6 ??com}

Page 3: optimized_filter_2_std_comp.lsp

Bass: L1 8.2 mH?{68 µF?com}?{0.15 mH+10 µF+1 ??com}?{6.8 ?+33 µF?com}

Mid: L1 0.56 mH?C1 68 µF?{C2 18 µF?com}?{L2 3.3 mH?com}?(22 ? ? 6.8 µF ? 0.15 mH)?{6.8 ?+15 µF?com}

Treble: C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{5.6 ??com}

Page 4: optimized_filter_3_std_comp.lsp

Bass: L1 8.2 mH?{68 µF?com}?{0.15 mH+10 µF+1 ??com}?{6.8 ?+33 µF?com}

Mid: L1 0.56 mH?C1 68 µF?{C2 18 µF?com}?{L2 3.3 mH?com}?(22 ? ? 6.8 µF ? 0.15 mH)?{6.8 ?+15 µF?com}

Treble: C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{4.7 ??com}

Notes

Topology is reconstructed from the coordinates stored in the .lsp files and redrawn; it reproduces LspCAD's own netlist, not its pixel layout.

Component values are exactly as stored (3 decimal digits trimmed). Drivers named Bass / Mid / Treble.

Mid and Treble are saved with inverted polarity in all three files; the schematics mark this on the driver symbol.

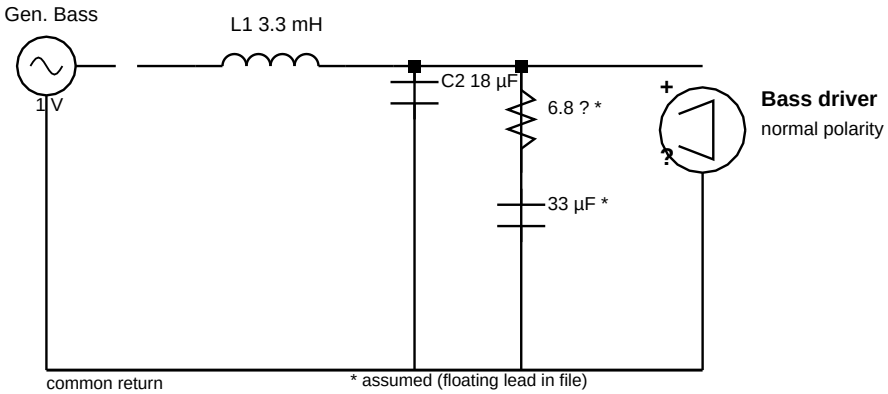
Each branch is driven from its own ideal 1 V generator (one per driver in the files), drawn here at each branch input.

?Kanske något.flt? is a binary LspCAD curve export without schematic information and is therefore not included.

In all three files the bass Zobel (6.8 ? + 33 µF) has one floating lead: its top terminal sits ~2.5 units from but not on any conductor. It is drawn attached to the bass hot node and marked ?assumed?.

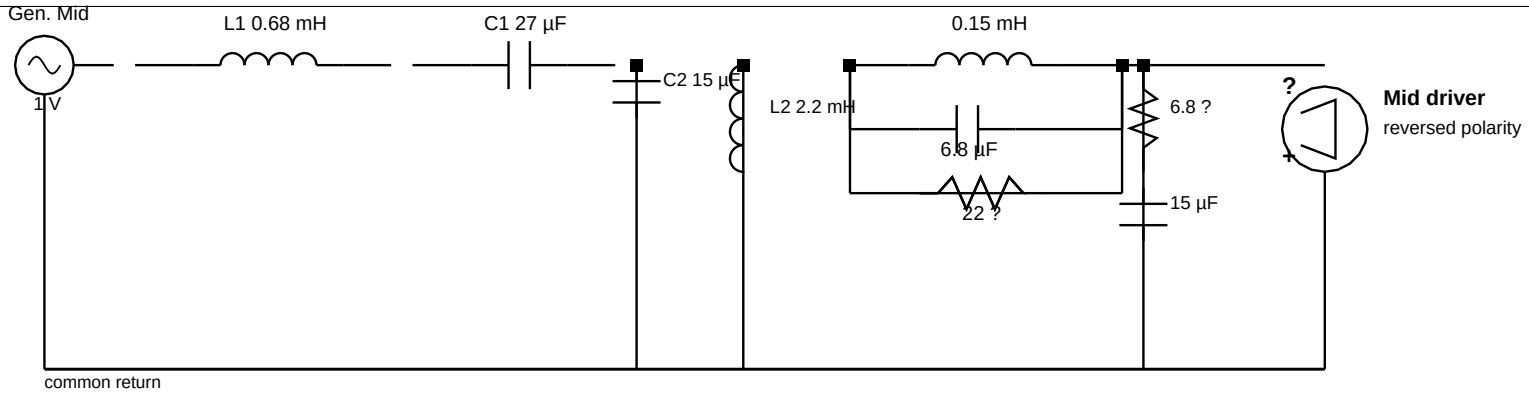
Optimized filter 1

optimized_filter_1_std_comp.lsp ? redrawn netlist (LspCAD 6 GenFilter)



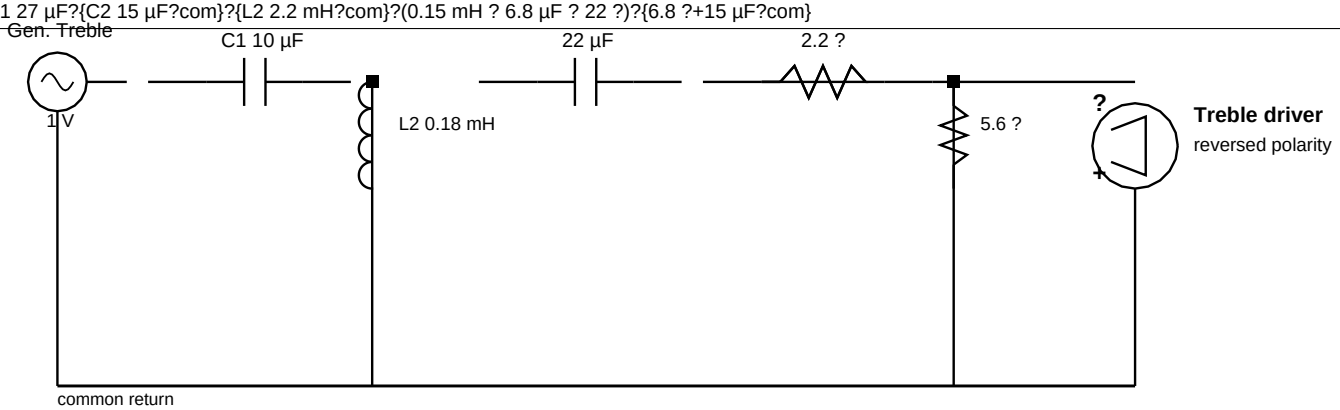
Bass branch ? Gen. Bass

L1 3.3 mH?{C2 18 µF?com}?{6.8 ?+33 µF?com}



Mid branch ? Gen. Mid

L1 0.68 mH?C1 27 µF?{C2 15 µF?com}?{L2 2.2 mH?com}?(0.15 mH ? 6.8 µF ? 22 ?)?{6.8 ?+15 µF?com}

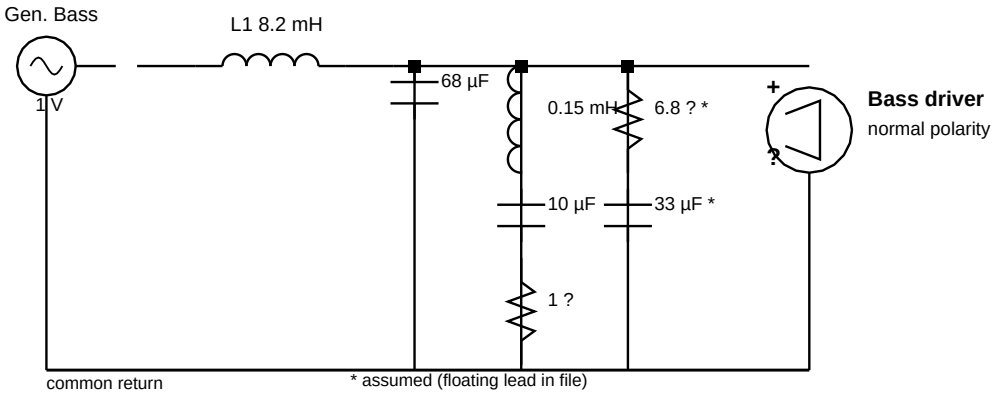


Treble branch ? Gen. Treble

C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{5.6 ??com}

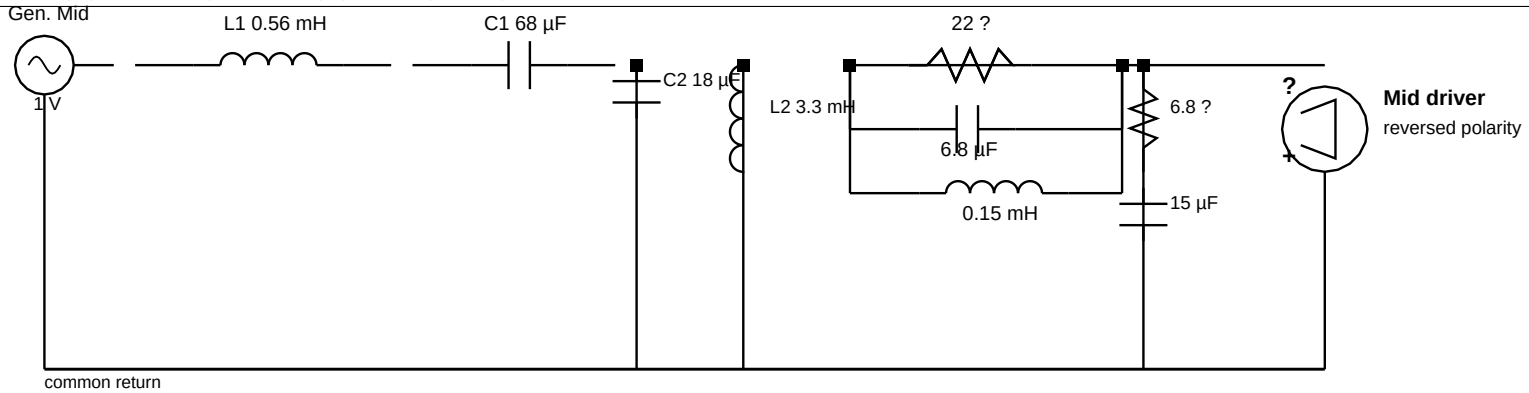
Optimized filter 2

optimized_filter_2_std_comp.lsp ? redrawn netlist (LspCAD 6 GenFilter)



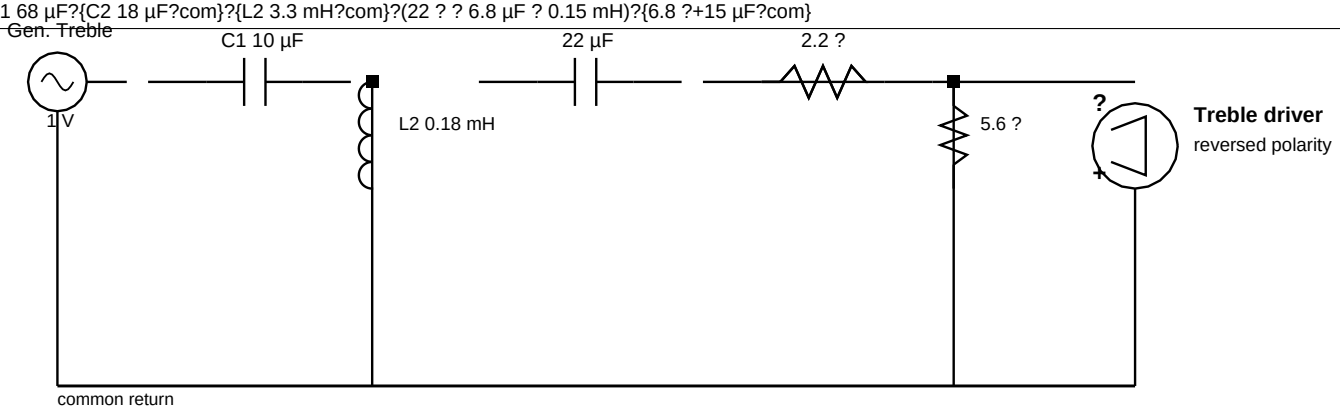
Bass branch ? Gen. Bass

L1 8.2 mH?{68 µF?com}?{0.15 mH+10 µF+1 ??com}?{6.8 ?+33 µF?com}



Mid branch ? Gen. Mid

L1 0.56 mH?C1 68 µF?{C2 18 µF?com}?{L2 3.3 mH?com}?(22 ? ? 6.8 µF ? 0.15 mH)?{6.8 ?+15 µF?com}

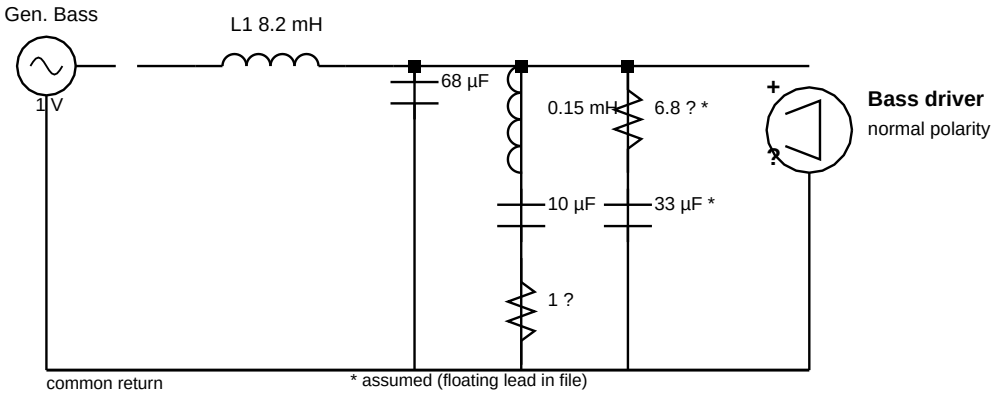


Treble branch ? Gen. Treble

C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{5.6 ??com}

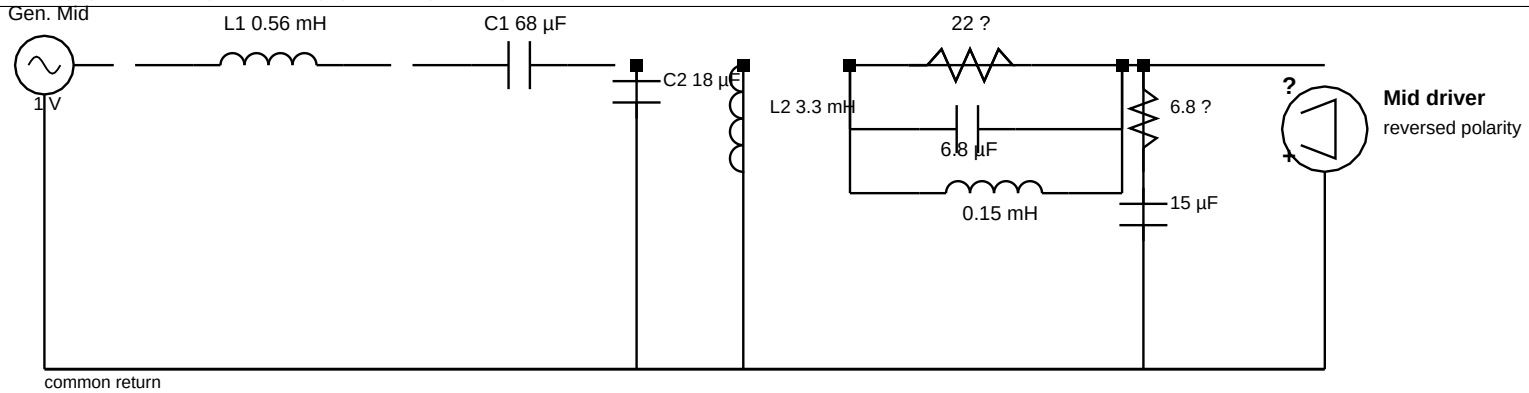
Optimized filter 3

optimized_filter_3_std_comp.lsp ? redrawn netlist (LspCAD 6 GenFilter)



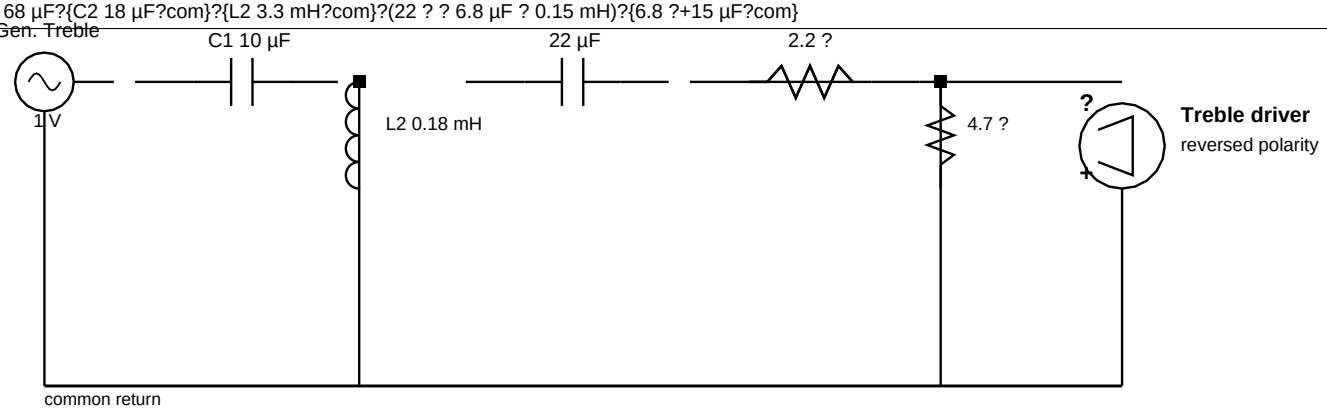
Bass branch ? Gen. Bass

L1 8.2 mH?{68 µF?com}?{0.15 mH+10 µF+1 ??com}?{6.8 ?+33 µF?com}



Mid branch ? Gen. Mid

L1 0.56 mH?C1 68 µF?{C2 18 µF?com}?{L2 3.3 mH?com}?(22 ? ? 6.8 µF ? 0.15 mH)?{6.8 ?+15 µF?com}



Treble branch ? Gen. Treble

C1 10 µF?{L2 0.18 mH?com}?22 µF?2.2 ??{4.7 ??com}